

Curation Re-evaluation Migration Spec

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Status: Design proposal

Scope: Preserve prior work as evidence, not as unquestioned truth

Decision summary

We should not discard prior AI characterization work.

We should also not promote prior curation outputs to ground truth under the new hierarchy model.

The correct approach is:

1. Keep raw source records and prior AI outputs.
2. Freeze legacy curation results as v1 outputs.
3. Re-run categorization under the new multi-axis model.
4. Treat prior Gemini analysis as evidence in the new system, not as final classification.

What survives unchanged

These remain first-class evidence and should not be thrown away:

Raw source and media evidence

From `assets` and related source metadata:

- `source`
- `source_ref`
- `source_url`
- `image_url`
- `stored_path`
- `thumb_path`
- `board`
- `description`
- `notes`
- `creator_name`
- `created_at`
- `imported_at`
- `scrape_json`
- media files on disk

These are closest to original acquisition context.

User scope and curation state

- `triage_status` (keeper, pending, etc.)
- collections / hidden state

- annotations

These are human workflow signals and should be preserved exactly.

Prior AI evidence

Keep all of the following:

- `asset_ai` rows for `provider='gemini'`
- `asset_ai` rows for `provider='gemini-video'`
- `asset_labels`
- `asset_embeddings`
- `ai_summary`

Important distinction:

- `asset_ai.json` and `asset_labels` are useful evidence.
- `ai_summary` is convenience text and should carry lower trust than structured evidence.

What should be demoted to legacy

These should be preserved, but not reused as authoritative truth for the new classifier:

Legacy curation outputs

- `style-best-of.json`
- `construction-concerns.json`
- `curation-manifest.json`
- HTML renders derived from those
- legacy dossier outputs from `Exploration/`

These are valuable for comparison and auditing, but they encode the old taxonomy assumptions.

Legacy inferred decisions

Do not treat the following as canonical training truth:

- old room assignment
- old concern type assignment
- old machine star ratings
- old pairwise ranking results
- old dossier summaries

Those outputs were downstream of the older room-centric logic and may reflect categorization errors.

What should be treated cautiously

Titles

Current `title` values are useful but not intrinsically trustworthy.

They may originate from:

- imported source titles
- Facebook post text truncation
- Houzz slug/title fallback logic
- AI scan title suggestion
- title audit application
- manual edit flows

Recommendation:

- keep current title as a display field
- add provenance for where the current title came from
- do not let title dominate classification unless other evidence agrees

AI summaries

Keep them, but use them as low-to-medium trust evidence only.

They are descriptive, but they are already lossy interpretations of the asset.

Trust model for reused evidence

High trust

- explicit user board intent when semantically meaningful
- human overrides
- `keeper` / `pending` triage state
- durable source-native structured metadata

Medium trust

- prior Gemini structured JSON fields such as:
 - `image_type`
 - `rooms`
 - `materials`
 - `styles`
 - `fixtures`
 - `appliances`
 - `brands_products`
- AI labels
- embeddings

Lower trust

- current title without provenance weighting
- AI prose summary
- heuristic expansions
- legacy curation assignment outputs

Recommendation on Gemini reuse

Do not blindly rerun everything from zero if we already have usable Gemini evidence.

Recommended policy:

1. Reuse prior generic Gemini image-analysis JSON as evidence.
2. Re-evaluate track and hierarchy membership using the new schema.
3. Do not reuse old Gemini curation decisions as final labels.

This means:

- keep the old image understanding
- replace the old taxonomy decision layer

Proposed new schema surface

Keep the existing evidence tables. Add a new classification layer instead of rewriting existing evidence in place.

1) classification_runs

Purpose:

- version each evaluation pass
- record schema and prompt/model versioning

Suggested fields:

- id
- schema_version
- run_type (track_gate, multi_axis_inference, human_override_merge, etc.)
- model_provider
- model_name
- prompt_version
- created_at
- notes

2) asset_field_provenance

Purpose:

- track where mutable display fields came from

Suggested fields:

- id
- asset_id
- field_name (title, description, board, ai_summary, etc.)
- origin_type (imported, source_native, ai_suggested, title_audit, manual_edit, derived)
- origin_ref
- actor
- confidence
- created_at
- superseded_at
- is_current

Minimum immediate use:

- backfill provenance for `title`

3) `asset_track_assessments`

Purpose:

- store track-level classification per run

Suggested fields:

- id
- run_id
- asset_id
- track
- confidence
- is_ambiguous
- decision_source (model, heuristic, merged, human_override)
- created_at

4) `asset_axis_memberships`

Purpose:

- store multi-valued memberships for the new hierarchy

This should be generic, not room-only.

Suggested fields:

- id
- run_id
- asset_id
- track
- axis_name

- axis_value
- confidence
- rank
- is_primary
- is_ambiguous
- created_at

Examples:

- axis_name='space_context', axis_value='outdoor_zone'
- axis_name='function', axis_value='dining'
- axis_name='product_focus', axis_value='sink'
- axis_name='concern_domain', axis_value='envelope'
- axis_name='product_system_focus', axis_value='zip_system'

This table is the core of the non-exclusive model.

5) asset_axis_evidence

Purpose:

- explain why a membership was assigned

Suggested fields:

- id
- run_id
- asset_id
- axis_name
- axis_value
- evidence_type (board, asset_ai_json, asset_label, title, description, embedding_neighbor, human_override)
- evidence_ref
- weight
- confidence
- note
- created_at

This is critical for auditability and reviewer trust.

6) asset_overrides

Purpose:

- apply human corrections without mutating the original machine evidence

Suggested fields:

- id
- asset_id
- axis_name

- `axis_value`
- `operation` (add, remove, set_primary, suppress)
- `actor`
- `note`
- `created_at`
- `expires_at` (nullable)

Migration policy by artifact

Keep as evidence

- `assets`
- `asset_ai`
- `asset_labels`
- `asset_embeddings`
- `title_audit_*`
- collections and triage state

Freeze as legacy comparison artifacts

- current curation JSON outputs
- HTML report outputs
- legacy Exploration dossier outputs

Recompute under new model

- `track`
- style hierarchy memberships
- construction hierarchy memberships
- room assignment
- product / system focus
- ambiguity flags
- export groupings derived from those

Practical migration steps

Phase 1: Snapshot legacy

- preserve current export directories as `v1`
- do not delete prior outputs

Phase 2: Add provenance and new classification tables

- add new tables through `db.py` migration path
- backfill `title` provenance first

Phase 3: Build v2 evaluator

Inputs:

- raw source fields
- prior Gemini structured JSON
- labels
- embeddings
- user scope / triage signals

Outputs:

- track assessment
- multi-axis memberships
- evidence rows
- ambiguity markers

Phase 4: Keep old and new side by side

- legacy outputs remain reviewable
- new outputs use v2 schema
- compare disagreement rates before replacing old export path

What not to do

- do not overwrite old evidence with new classification outputs
- do not train the new system on old room assignments as if they were ground truth
- do not let pairwise ranking compensate for bad categorization
- do not force a single room when `space_context`, `subject_type`, or `product_focus` suggests otherwise

Minimal viable implementation order

If we want the smallest defensible implementation:

1. add `classification_runs`
2. add `asset_field_provenance`
3. add `asset_track_assessments`
4. add `asset_axis_memberships`
5. backfill title provenance
6. build new track gate
7. build new multi-axis membership inference

`asset_axis_evidence` should come immediately after if we care about explainability, which we probably should.

Final recommendation

We should keep most of the previous attempts, but recast them:

- raw source data stays authoritative
- generic Gemini visual characterization stays as reusable evidence
- legacy curation decisions become historical comparison artifacts
- the new agent should evaluate classification anew, but using prior evidence rather than ignoring it

That is the safest path that preserves past work without inheriting its assumptions.